*[ LABORATORY LOGO ]*

# [LABORATORY NAME]

## STANDARD OPERATING PROCEDURE

# Venous Blood Sample Collection by Venipuncture

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## Approval and Authorisation

| Role | Name (print) | Signature | Date |
|---|---|---|---|
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| **Reviewed by** | [Name — Laboratory Manager] | | |
| **Approved by** | [Name — Laboratory Director] | | |

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## Revision History

| Version | Date | Description of change | Prepared | Approved |
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- Controlled copies: Phlebotomy station; Sample reception; Laboratory Manager’s office; each collection point.
- A read-only copy is published on the laboratory intranet. Printed copies are uncontrolled unless stamped “CONTROLLED COPY” in red and listed in the distribution register (ISO 15189:2022, 8.3).

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## 1. Purpose

This Standard Operating Procedure (SOP) defines the standard method for collecting venous blood samples by peripheral venipuncture at [LABORATORY NAME]. Following this procedure protects patient safety, guarantees positive patient identification and sample integrity, and supports reliable examination results in accordance with ISO 15189:2022, in particular the pre-examination process requirements of Clause 7.2 and the personnel competence requirements of Clause 6.2.

## 2. Scope

This SOP applies to all personnel (employees, students and contractors) who collect venous blood samples at the main laboratory and at every collection point operated by the laboratory, and to all patient populations served.

Out of scope — covered by separate procedures:

- arterial blood collection (SOP-PRE-004);
- capillary blood sampling (SOP-PRE-005);
- blood culture collection (SOP-PRE-002);
- blood donation and donor phlebotomy (SOP-BB-001).

## 3. Definitions and Abbreviations

| Term | Definition |
|---|---|
| Positive patient identification | Verification of patient identity using at least two unique identifiers (full name and date of birth or national identification number) matched to the request. |
| Order of draw | The standard sequence in which blood collection tubes are filled during a single venipuncture, preventing additive cross-contamination between tubes. |
| Haemolysis | Rupture of red blood cells releasing intracellular constituents into plasma or serum; may invalidate results and is a cause for sample rejection. |
| Mislabelled sample | A sample whose label does not correctly and uniquely identify the patient from whom it was collected; always treated as nonconforming work (ISO 15189:2022, 7.5). |
| EDTA / LIS / PPE / TAT / NC / IQC | Ethylenediaminetetraacetic acid (tube anticoagulant) / laboratory information system / personal protective equipment / turnaround time / nonconforming (work) / internal quality control. |

## 4. Responsibilities

| Role | Responsibility |
|---|---|
| Phlebotomist / sample collector | Perform venipuncture exactly as written in this SOP; identify every patient with two identifiers; label samples at the point of collection; report and record incidents and nonconformities. |
| Reception / clerical staff | Verify completeness of the request form; register samples in the LIS; protect patient confidentiality (ISO 15189:2022, 4.2). |
| Laboratory Manager | Ensure sufficient trained and authorised personnel; formally authorise staff to perform venipuncture after competence assessment (6.2); review quality indicators. |
| Quality Manager | Maintain this SOP under document control (8.3); monitor sample rejection and haemolysis indicators; coordinate corrective actions (8.7); schedule periodic review. |
| Health & Safety Officer | Maintain sharps disposal infrastructure and PPE availability; investigate exposure incidents; oversee hepatitis B vaccination of collectors. |

## 5. Safety and Precautions

- Apply standard precautions for every patient regardless of diagnosis: hand hygiene before and after every collection, gloves for every procedure, gloves changed between patients.
- Wear a protective gown and eye protection when splashing of blood is anticipated.
- Never recap needles. Activate the needle safety device immediately after withdrawal and dispose of the complete assembly in a puncture-resistant sharps container at the point of use.
- Hepatitis B vaccination is mandatory for all personnel who perform venipuncture.

- Needlestick or mucous-membrane exposure: wash or irrigate immediately, report at once to the Health & Safety Officer and follow SOP-SAF-003. Record the event; raise a nonconformity if procedural failure contributed (ISO 15189:2022, 7.5).
- Keep all patient information confidential at all times (ISO 15189:2022, 4.2).

## 6. Materials and Equipment

- Disposable vacuum blood collection tubes with approved additives: serum (plain red or gold top), sodium citrate 3.2% (light blue), EDTA (lavender), lithium heparin (green), fluoride/oxalate (grey).
- Safety-engineered needles 21G or 22G; butterfly sets for difficult veins and for paediatric patients.
- 70% isopropyl alcohol swabs, dry sterile gauze, adhesive plaster.
- Disposable single-patient tourniquets.
- Pre-printed barcode labels and/or LIS request forms; ballpoint pen for manual labels.
- Biohazard transport bags, rigid transport containers and cooled carriers where required.
- Puncture-resistant sharps containers and clinical waste bags.

## 7. Procedure

### 7.1 Patient preparation and identification

- Confirm the request: tests requested, patient details, fasting status and any special timing requirements.
- Identify the patient using two unique identifiers — full name plus date of birth or national identification number — matched against the request form and, where available, the patient's wristband. Ask inpatients to state their own name.
- Explain the procedure, confirm consent and position the patient comfortably, seated or lying down. Never leave a patient unattended in a collection chair.
- Perform hand hygiene and put on gloves.

### 7.2 Order of draw

When more than one tube is collected during a single venipuncture, fill the tubes in the sequence below (consistent with CLSI GP41). Invert each filled tube gently 8 to 10 times; never shake. Fill citrate and EDTA tubes to the marked fill volume.

| Order | Tube cap colour / additive | Typical examinations |
|---|---|---|
| 1 | Blood culture bottles (yellow) | Blood cultures |
| 2 | Light blue — sodium citrate 3.2% | Coagulation (PT, aPTT, INR) |
| 3 | Red / gold — plain or clot activator (with gel) | Clinical chemistry, serology, immunology |
| 4 | Green — lithium heparin | Plasma chemistry, emergency tests |
| 5 | Lavender — EDTA | Haematology (CBC), HbA1c |
| 6 | Grey — fluoride / oxalate | Glucose, lactate |

### 7.3 Venipuncture

1. Assemble all required materials within reach before approaching the patient.
2. Select the site: prefer the median cubital vein of the forearm. Avoid the arm on the side of a mastectomy, arteriovenous fistula, active infusion, burn or scar.
3. Apply the tourniquet 3 to 4 cm above the intended puncture site, tight enough to impede venous return but not arterial flow. Do not leave the tourniquet on for more than 1 minute.
4. Palpate the vein with the index finger. Clean the site with a 70% isopropyl alcohol swab using a circular motion from the centre outwards over at least 2 cm and allow to air-dry for at least 30 seconds. Do not touch, blow on or fan the site.
5. Anchor the vein with the thumb of the non-dominant hand, holding the skin taut below the puncture point.
6. Insert the needle with the bevel upwards at an angle of 15 to 30 degrees, in a single smooth motion, in the direction of the vein.
7. When blood flows, fill the tubes in the order of draw given in 7.2. Under-filling or over-filling citrate and EDTA tubes is a cause for rejection.

8. Release the tourniquet before withdrawing the needle. Withdraw the needle and apply dry gauze with firm pressure for 2 to 3 minutes; apply plaster if needed.
9. Activate the needle safety device immediately and drop the entire unit into the sharps container at the point of use — never carry an unsheathed needle and never recap.
10. Keep the patient's arm extended and observe for haematoma or dizziness before the patient leaves the collection area.
11. If blood does not flow or flows slowly: do not probe more than twice. Release the tourniquet, withdraw, apply pressure, and use a second site or the other arm. Record every difficult draw.

### 7.4 Labelling and documentation

- Label every tube at the bedside, at the point of collection, immediately after filling. Never pre-label tubes and never label tubes away from the patient.
- Minimum label data: patient full name; unique identifier (date of birth or national ID / LIS number); collection date and exact time; collector's initials.
- Verify that every label matches the request form before leaving the patient. Place the label lengthwise on the tube without covering the cap.
- For manual labels, write legibly with a ballpoint pen. Correct errors with a single line through the error, the correct data, an initial and the date (ISO 15189:2022, 8.4).

### 7.5 Transport to the laboratory

- Place tubes upright in a biohazard bag; keep the request form from direct contact with the tubes.
- Deliver to the laboratory within 2 hours of collection, at 18 to 25 degrees C unless otherwise specified (for example cooled carriers at 2 to 8 degrees C for glucose and lactate).
- Do not use the pneumatic tube system for coagulation samples or blood culture bottles unless the system has been validated for them.

### 7.6 Sample acceptance and rejection criteria

Samples are checked at reception against the criteria below. Rejected samples are recorded in the Sample Rejection Register and, where the cause is a systemic failure, handled as nonconforming work (ISO 15189:2022, 7.5).

| Observation | Action |
|---|---|
| Patient identification missing, incomplete, or not matching the request form | Reject; notify the requester; arrange recollection; record as mislabelling NC |
| Visible haemolysis in plasma/serum, or analyser haemolysis flag | Reject the affected test(s); notify; recollect |
| Incorrect tube or additive for the test requested | Reject the affected test(s); recollect |
| Insufficient volume (especially citrate and EDTA tubes) | Reject the affected test(s); recollect |
| Clots present in an EDTA tube (haematology) | Reject; recollect immediately |
| Transport time or temperature outside defined limits | Assess analyte stability; if stability is not demonstrated, reject and recollect |
| Damaged or leaking container | Reject; recollect; decontaminate the transport container |

## 8. Quality Control and Monitoring

- Quality indicators monitored monthly by the Quality Manager: sample rejection rate (target at most 2% of all samples); haemolysis rate (target at most 1%); mislabelled samples (target zero); samples delivered within the defined turnaround time.
- Indicator results are reported at the monthly quality meeting and reviewed in management review (ISO 15189:2022, 8.8 and 8.9).
- Where available, the laboratory participates in external quality assessment schemes for pre-analytical phases and acts on unsatisfactory performance (8.7).
- Phlebotomy technique is observed directly at least annually for each collector as part of competence assessment (6.2).

## 9. Risk Considerations

Risks relevant to this procedure are recorded below in line with the risk-based approach of ISO 15189:2022 (Clauses 5.6 and 8.5). Residual risks are reviewed at each periodic review of this SOP.

| Risk | Control measure |
|---|---|
| Wrong-patient identification | Two identifiers confirmed at the bedside; labelling at the point of collection; labels verified against the request before leaving the patient |
| Needlestick injury | Safety-engineered devices; no recapping; sharps container at the point of use; mandatory hepatitis B vaccination; SOP-SAF-003 for exposure management |
| Haemolysed or clotted samples | Correct tube fill volumes; gentle inversion; order of draw; avoid prolonged tourniquet time; prompt transport |
| Delayed transport affecting analyte stability | Two-hour delivery rule; scheduled courier runs; temperature-controlled carriers |

## 10. Documentation and Records

- Records generated by this procedure: blood collection register / LIS collection log; sample rejection register (F-PRE-002); exposure incident reports (F-SAF-001); the training acknowledgement in Annex A (F-TRN-001).
- Records are created at the time the activity is performed, and remain legible, traceable to the collector and retrievable (ISO 15189:2022, 8.4).
- Corrections: single line through the error, the correct value, an initial and the date. The original entry must remain legible.
- Retention: at least 5 years, or as required by national regulation where longer; disposal by approved confidential destruction.

## 11. References

- ISO 15189:2022 — Medical laboratories: requirements for quality and competence (in particular Clauses 4.2, 4.3, 5.6, 6.2, 7.2, 7.5, 8.3, 8.4, 8.5).
- CLSI GP41 — Collection of Diagnostic Venous Blood Specimens.
- CLSI H3 — Procedures for the Handling and Processing of Blood Specimens.
- WHO — Laboratory Quality Management System (LQMS) Handbook.
- [National phlebotomy, sharps-safety and infection-prevention guidelines — adapt to the applicable national authority].

## 12. Related Documents and Forms

| Reference | Title |
|---|---|
| SOP-PRE-002 | Sample Reception, Accessioning and Rejection |
| SOP-SAF-003 | Occupational Exposure and Post-Exposure Management |
| SOP-QM-004 | Control of Documents and Records |
| F-PRE-001 | Blood Collection Register |
| F-PRE-002 | Sample Rejection Register / Nonconforming Work Form |
| F-TRN-001 | Training Acknowledgement and Competence Record (Annex A) |

## Annex A — Training and Competence Acknowledgement (Form F-TRN-001)

I confirm that I have received training on this Standard Operating Procedure, that I have read and understood its contents including the safety precautions, and that I agree to perform the procedure as written. I understand that I must not perform this procedure until this form has been signed and filed in my training record. Training, competence assessment and authorisation of personnel are maintained in accordance with ISO 15189:2022, Clause 6.2; the completed form is retained as a record in accordance with Clause 8.4.

| S/No | Full Name (print) | Staff / Employee ID | Designation | Signature | Date |
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Trainer / Supervisor: _____ Name (print): _____ Date: _____

*Complete one row per staff member (15 rows provided — at least 12 anticipated). If more rows are needed, attach continuation sheet F-TRN-001-2 and staple it to this page. File the completed form in the personnel training record; reference it in the SOP training register maintained by the Quality Manager.*